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***STATISTICS PROJECT***

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# **STATISTICS PROJECT REPORT**

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Does higher education enrollment reduce income inequality in Armenia?

# INTRODUCTION

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Income inequality has been a long-standing problem in Armenia. Although the government has introduced educational reforms, we do not know if more education has helped reduce inequality. The goal of this project is to explore whether **higher education enrollment reduces income inequality** in Armenia. Specifically, we want to answer: Does higher enrollment in secondary, bachelor, or master programs lower the GINI inequality index?

## DATA DESCRIPTION

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We used three datasets:

- **Income data:** average national income per year
- **GINI coefficient data:** standard measure of income inequality (0 = equal, 100 = extreme inequality)
- **Education data:** secondary, bachelor, and master enrollment rates (%)

**Variables:**

| Variable             | Description                       | Unit  |
|----------------------|-----------------------------------|-------|
| Year                 | Calendar year                     | Year  |
| Absolute income      | Average monthly income            | AMD   |
| GINI coefficient     | Income inequality index           | 0–100 |
| Secondary enrollment | % enrolled in secondary school    | %     |
| Bachelor enrollment  | % enrolled in bachelor's programs | %     |
| Master enrollment    | % enrolled in master's programs   | %     |

**Sample size:**

- Income-GINI: ~20 years
- Education-GINI: 2015–2017
- Missing values were removed to keep datasets clean.

## EXPLORATORY DATA ANALYSIS (EDA)

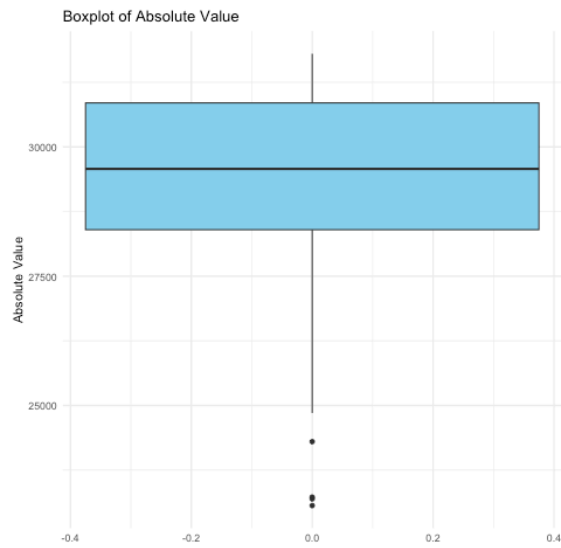
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The purpose of this step was to **explore patterns, detect outliers, and understand distributions.**

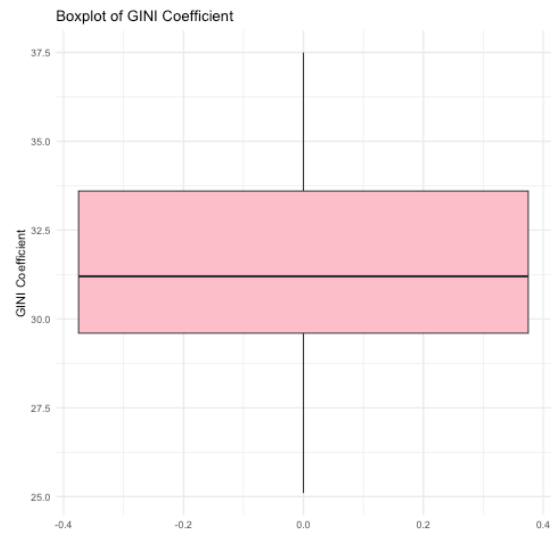
**Summary statistics**

- Income: ~180,000 AMD on average, wide range from ~90,000 to ~333,000 AMD
- GINI: ~30.5 on average, range 25–37

## Boxplots

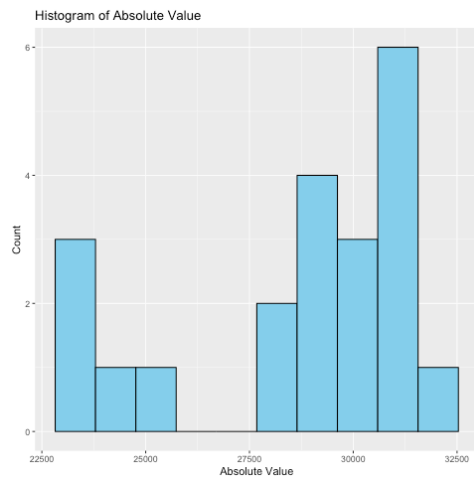


shows wide income variation,  
some outliers

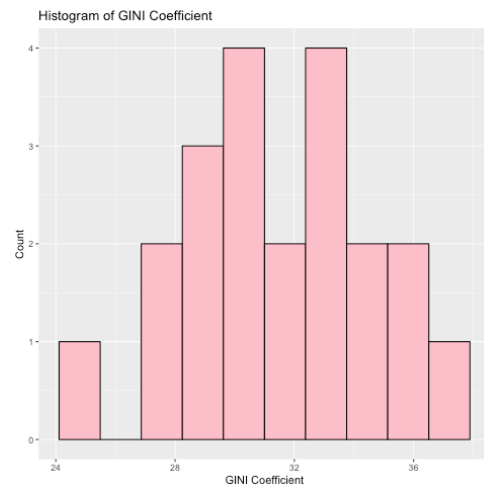


stable GINI without  
extreme value

## Histograms

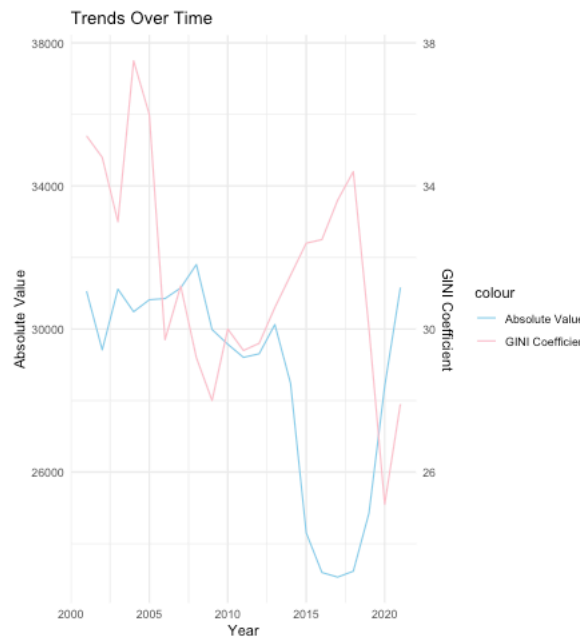


shows two income clusters  
(bimodal)



shows most GINI  
values around ~28–33

## Time trends



income steadily rises;

GINI fluctuates slightly  
downward

## Conclusion:

Income increased over time, but inequality showed no clear decline. Education enrollment rates are stable or slightly increasing.

# METHODOLOGY

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**Goal:** Test if income or education enrollment can explain changes in inequality.

**Tests used:**

- Pearson correlation → check basic linear relationships
- Linear regression → test predictive effects of income or education on inequality

**Assumptions checked:**

- Numeric continuous data
- Linear relationships
- No extreme outliers after cleaning

**Software:**

- R, with tidyverse, ggplot2, broom, openxlsx packages

# RESULTS

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## Income vs. GINI

- **Correlation:**  
 $r = -0.13$ ,  $p = 0.57 \rightarrow$  weak, non-significant
- **Regression:**  
Slope  $\approx 0$ ,  $p = 0.57 \rightarrow$  no predictive effect

## Conclusion:

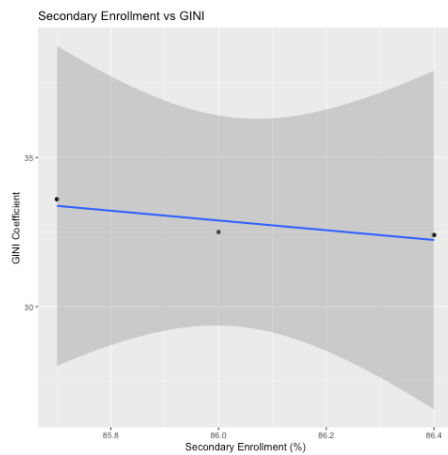
Rising income over the years did **not** reduce inequality in Armenia.

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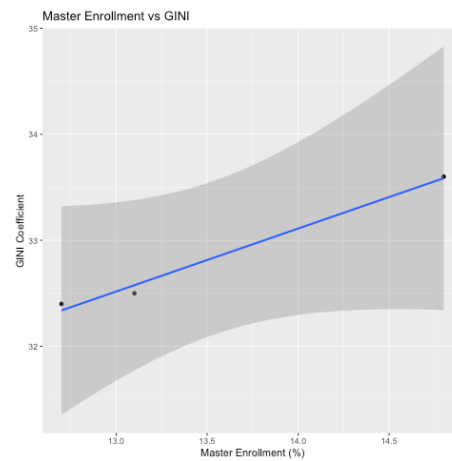
## Education vs. GINI correlations

- **Secondary enrollment:**  $r = -0.86$ ,  $p = 0.34 \rightarrow$  strong negative, not significant
- **Bachelor enrollment:**  $r = 0.96$ ,  $p = 0.17 \rightarrow$  strong positive, not significant
- **Master enrollment:**  $r = 0.99$ ,  $p = 0.07 \rightarrow$  very strong positive, nearly significant

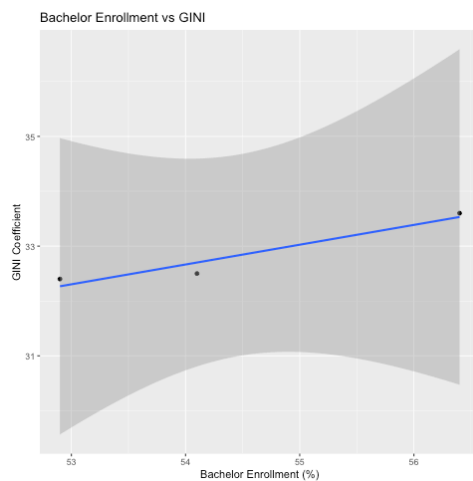
## Scatterplots



secondary enrollment vs. GINI  
(negative slope)



master enrollment vs. GINI  
(strong positive slope)



bachelor enrollment vs. GINI  
(positive slope)

## Conclusion:

None of the education levels significantly predicted lower inequality.

Surprisingly, bachelor's and master's enrollments showed positive links with inequality, but this may reflect small sample size (3 years) or other unmeasured factors.



# DISCUSSION

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## **Answer to research question:**

We found **no strong evidence** that more education reduces income inequality in Armenia.

## **Interpretation:**

- Income growth has **not** led to more equality.
- Secondary education showed a negative relationship with inequality, but it was not statistically significant.
- Bachelor and master enrollments showed positive associations with inequality, but these results need to be interpreted cautiously due to the very small dataset.

## **Practical meaning:**

Education alone may not fix inequality. Policymakers need to look beyond enrollment numbers and focus on job access, wages, and social programs.

## **Limitations:**

- Only three years of education data
- Small sample reduces statistical power
- Correlation analysis cannot prove causation
- Other social and economic factors were not included

# CONCLUSION

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Based on the available data, we conclude that neither income growth nor higher education enrollment has significantly reduced income inequality in Armenia. Future studies should use longer datasets and include more variables like employment rates and wages.

# REFERENCES

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